

Two Pointers Level - 1 Problems

1. Reverse an Array Pattern: Left + Right

```
arr = [1,2,3,4,5]

def rev_arr(arr):

    left = 0
    right = len(arr) - 1

    while left < right:

        arr[left], arr[right] = arr[right], arr[left]

        left += 1
        right -= 1

    return arr
print(rev_arr(arr))
```

2. Reverse a String Pattern: Left + Right

```
strs = "hari"

def rev_str(strs):

    strs = list(strs)

    left = 0
    right = len(strs) - 1

    while left < right:

        strs[left], strs[right] = strs[right], strs[left]

        left += 1
        right -= 1

    return "".join(strs)

print(rev_str(strs))
```

3. Check Palindrome Pattern: Left ↔ Right comparison

```
s = "level"

def is_palindrome(s):

    left = 0
    right = len(s) - 1

    while left < right:

        if s[left] != s[right]:

            return False

        left += 1
        right -= 1

    return True

print(is_palindrome(s))
```

4. Check Valid Palindrome String Pattern: Opposite Pointers

```
s = "A man, a plan, a canal: Panama"
# s = "race a car"

def is_valid_palindrome(s):

    left = 0
    right = len(s) - 1

    while left < right:

        if not s[left].isalnum():

            left += 1

        elif not s[right].isalnum():

            right -= 1

        elif s[left].lower() != s[right].lower():

            return False

        else:

            left += 1
            right -= 1

    return True

print(is_valid_palindrome(s))
```

5. Pair Sum in Sorted Array

Pattern: Left + Right

```
nums = [1, 2, 3, 4, 6]
target = 6

def pair_sum(nums, target):

    left = 0
    right = len(nums) - 1

    while left < right:

        total = nums[left] + nums[right]

        if total == target:
            return True

        elif total < target:
            left += 1

        else:
            right -= 1

    return False

print(pair_sum(nums, target))
```

6. Merge Two Sorted Arrays — Basic

Pattern: Two pointers

```
a = [1,3,5]
b = [2,4,6]

def merge_sorted_arr(a,b):
    i = j = 0

    result = []

    while i < len(a) and j < len(b):
        if a[i] <= b[j]:
            result.append(a[i])
            i += 1

        else:
            result.append(b[j])
            j += 1

    while i < len(a):
        result.append(a[i])
        i += 1

    while j < len(b):
        result.append(b[j])
        j += 1

    return result

print(merge_sorted_arr(a,b))
```